UML RELATIONSHIPS

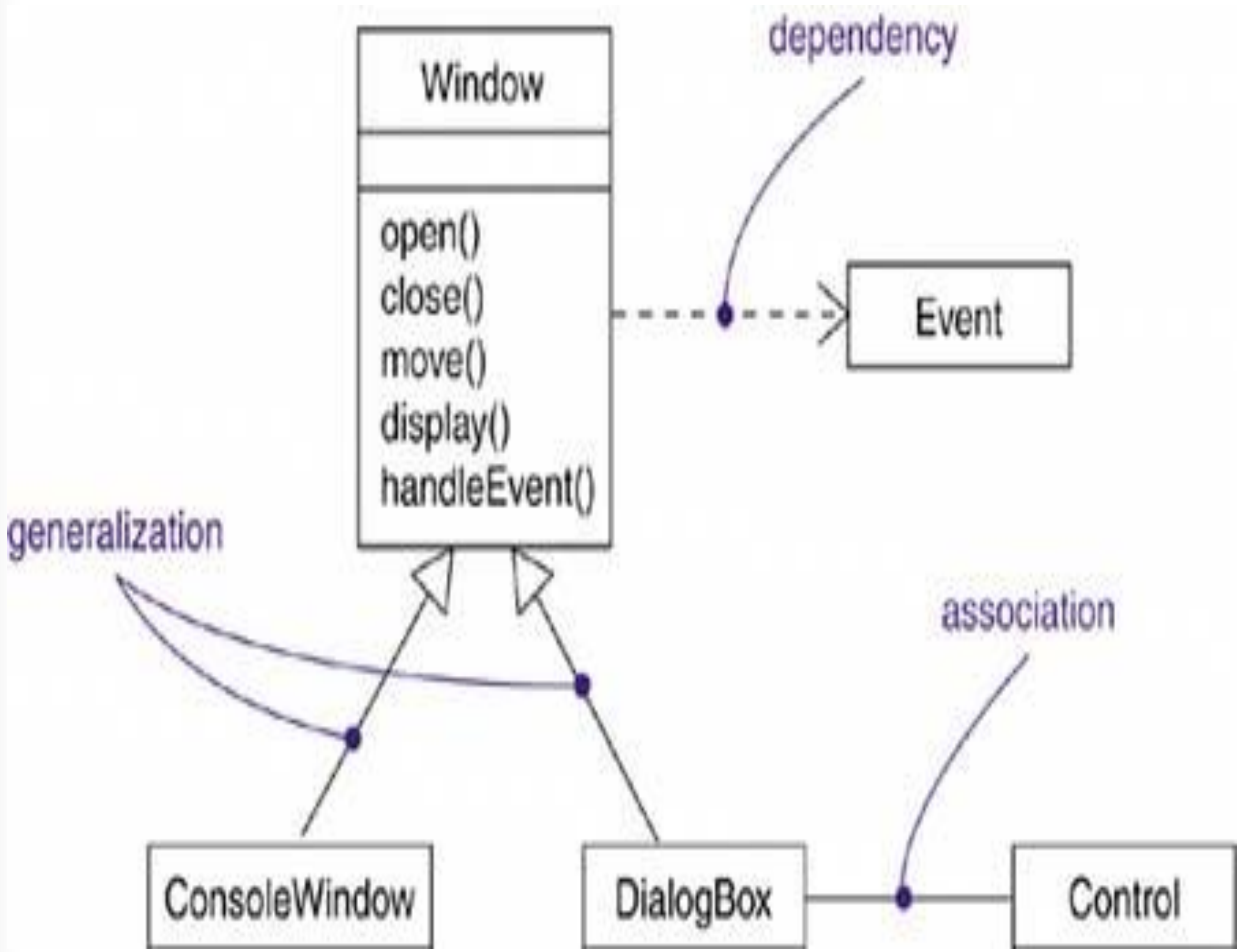
Relationship

- Is a connection among things
- Graphically represented as path, with different kinds of lines, used to distinguish the kinds of relationships.

Types of Relationships

- In the UML, the ways that things can connect to one another, either logically or physically, are modeled as relationships.
- In object-oriented modeling, there are three kinds of relationships that are most important:
- dependencies,
- generalizations,
- and associations.

Relationships

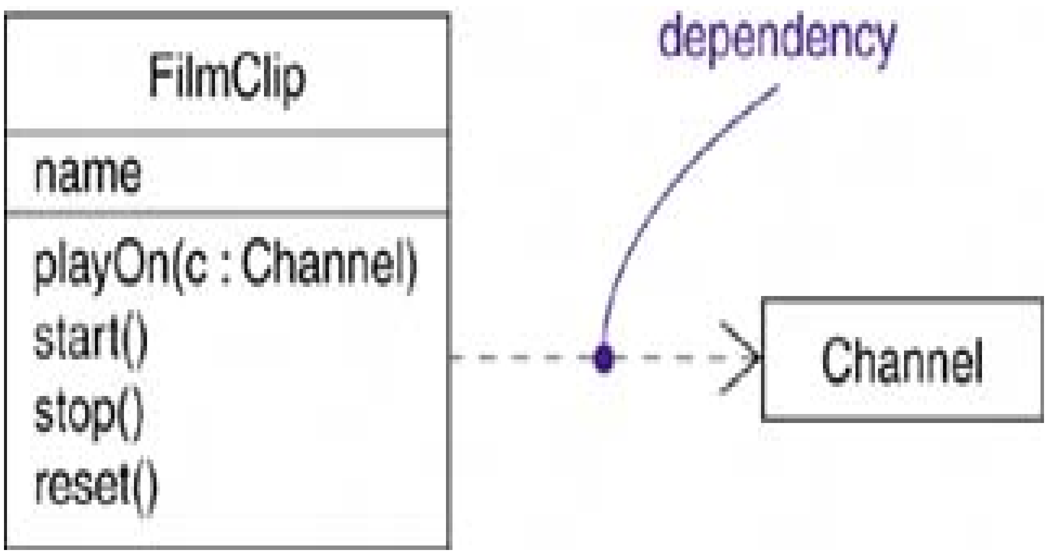


Dependency

- Definition: A *dependency* is a relationship that states that one thing (for example, class Window) uses the information and services of another thing (for example, class Event), but not necessarily the reverse.

- Graphically, a dependency is rendered as a dashed directed line, directed to the thing being depended on. Choose dependencies when you want to show one thing using another.
- Notation: ---------->

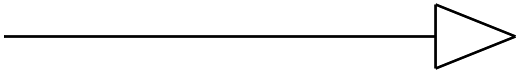
- Example:


FilmClip
name
playOn(c : Channel)
start()
stop()
reset()
dependency
Channel

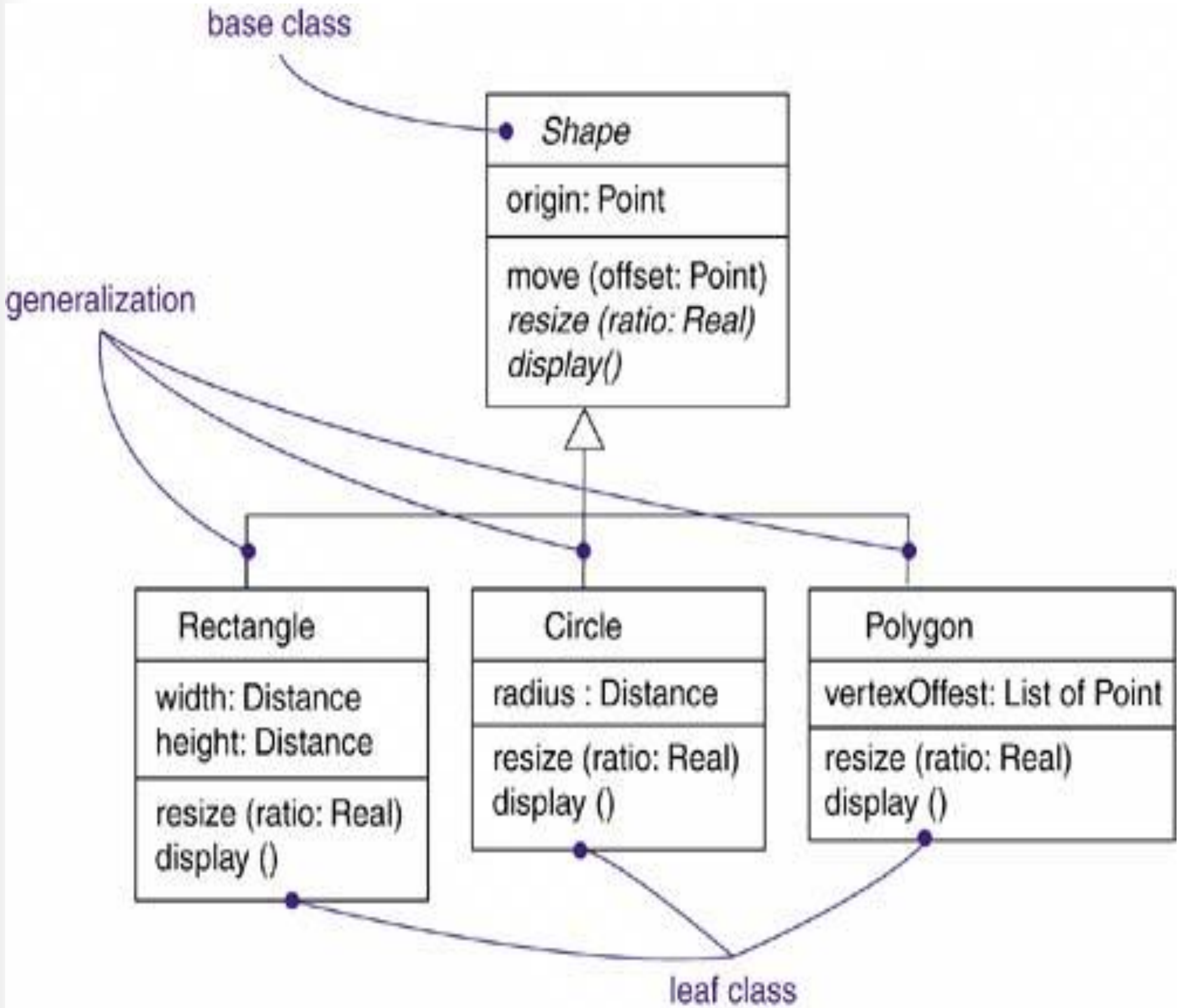
Generalization

- Definition: A *generalization* is a relationship between a general kind of thing (called the superclass or parent) and a more specific kind of thing (called the subclass or child).

- Generalization is sometimes called an "is-a-kind-of" relationship.
- Graphically, generalization is rendered as a solid directed line with a large unfilled triangular arrowhead, pointing to the parent.



Example:


base class
Shape
origin: Point
move (offset: Point)
resize (ratio: Real)
display()
generalization
Rectangle
width: Distance
height: Distance
resize (ratio: Real)
display ()
Circle
radius : Distance
resize (ratio: Real)
display ()
Polygon
vertexOffest: List of Point
resize (ratio: Real)
display ()
leaf class

Association

- Definition: An *association* is a structural relationship that specifies that objects of one thing are connected to objects of another.

- Given an association connecting two classes, you can relate objects of one class to objects of the other class.
- An association that connects exactly two classes is called a binary association.

- Graphically, an association is rendered as a solid line connecting the same or different classes. Use associations when you want to show structural relationships.
- Notation:

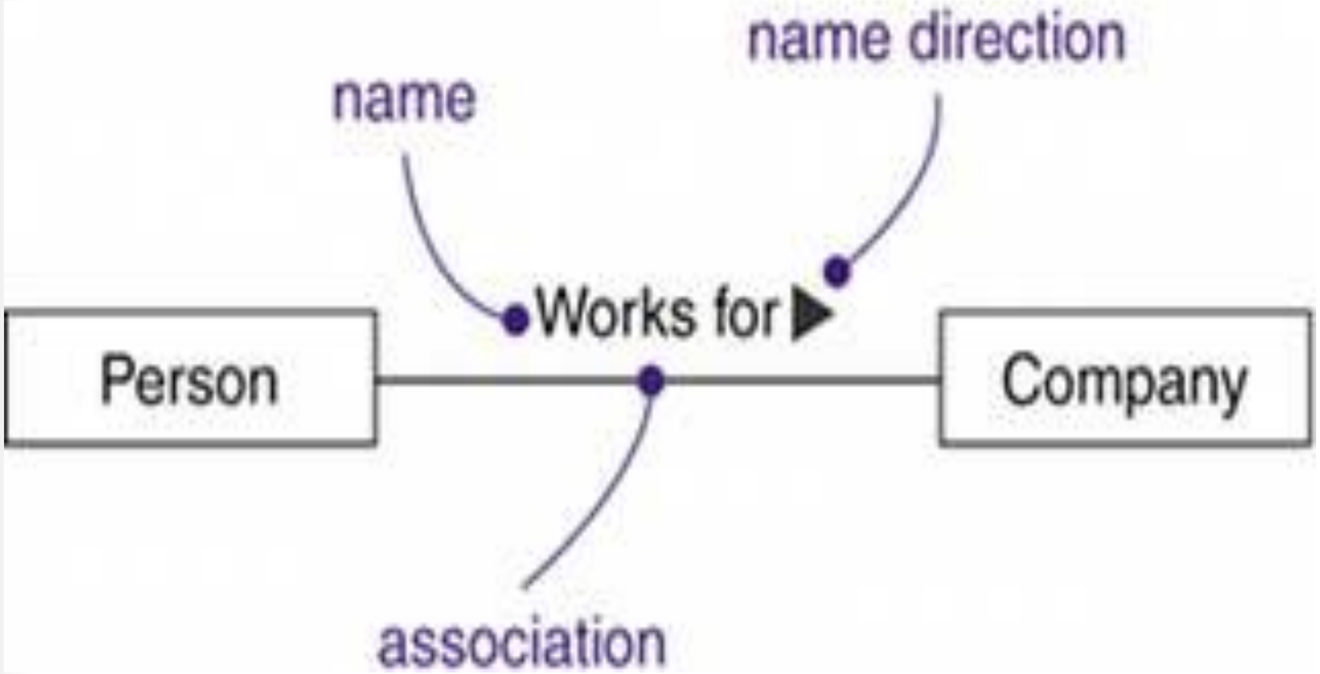
Four adornments that apply to associations:

- **Name:**

  An association can have a name, and you use that name to describe the nature of the relationship.

  You can give a direction to the name by providing a direction triangle that points in the direction you intend to read the name.

Example:

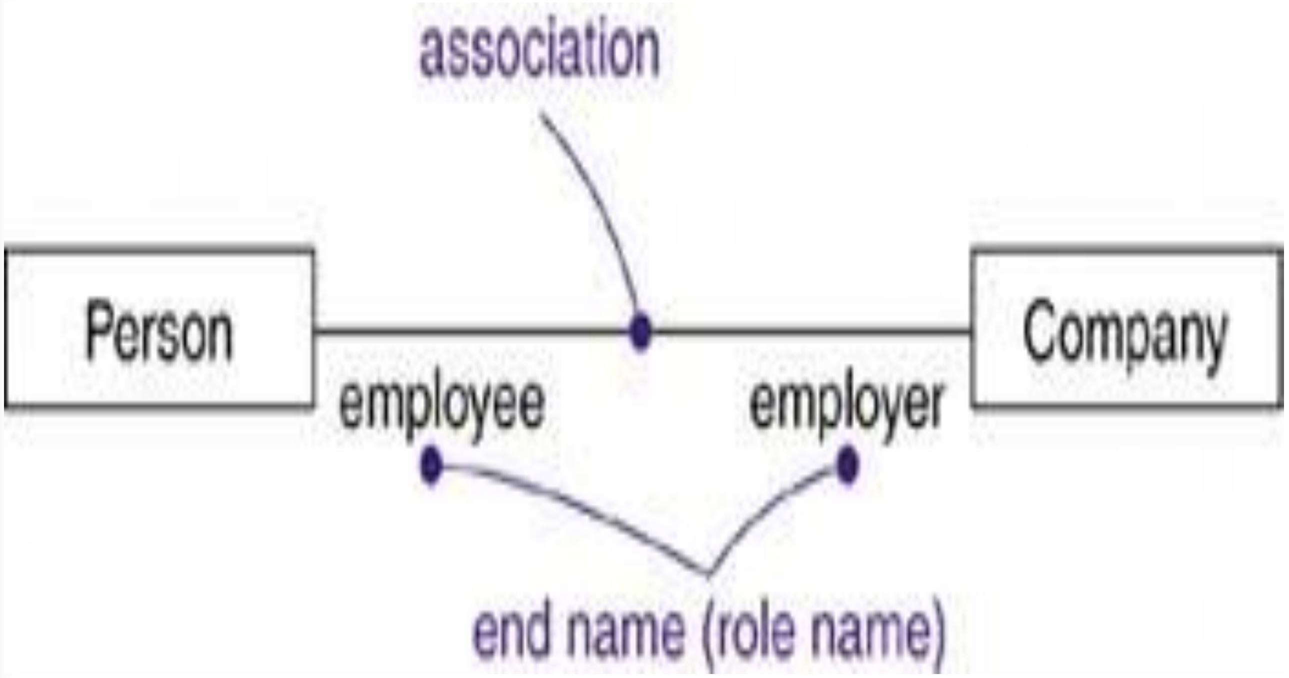


- **Role**

  A role is just the face the class at the far end of the association presents to the class at the near end of the association.

The role played by an end of an association is called an end name.

Examples:

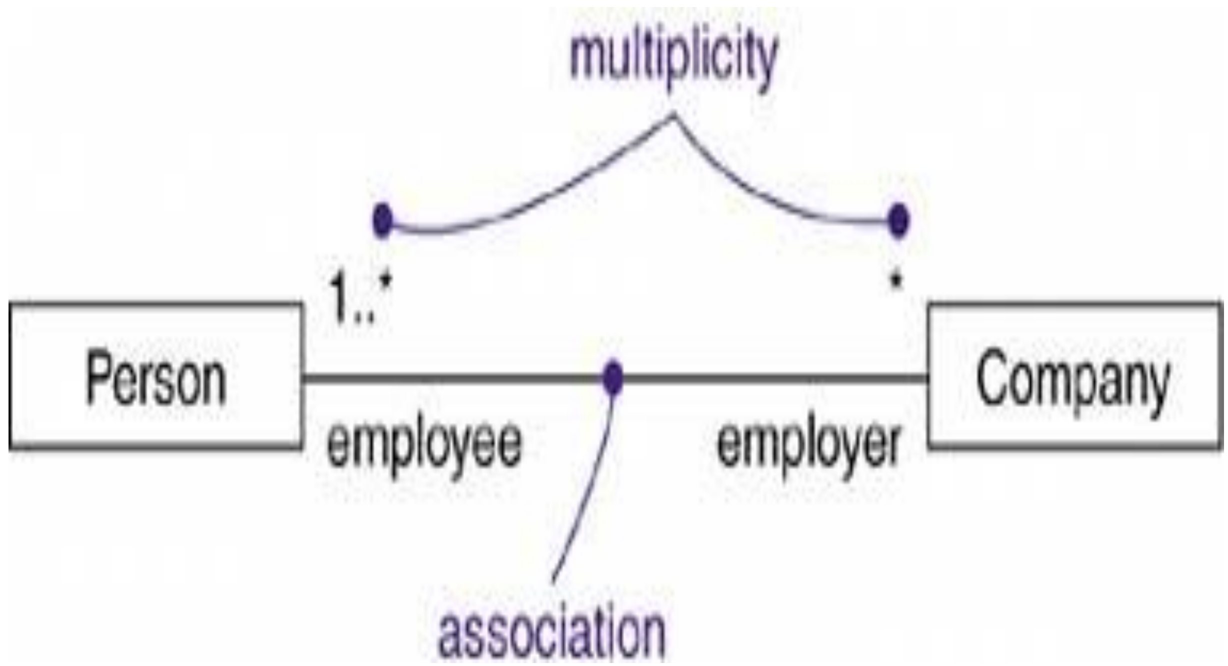

association
Person
Company
employee
employer
end name (role name)

Multiplicity

- Definition: Specifies how many instances of one class may relate to a single instance of an associated class.
- When you state a multiplicity at the one end of an association, you are specifying that, for each object of the class at the opposite end, how many objects at the near end may exist.

- You can show a multiplicity of exactly one (1),
- zero or one (0..1),
- many (0..*),
- or one or more (1..*).
- You can give an integer range (such as 2..5).
- You can even state an exact number (for example, 3, which is equivalent to 3..3).

Example:

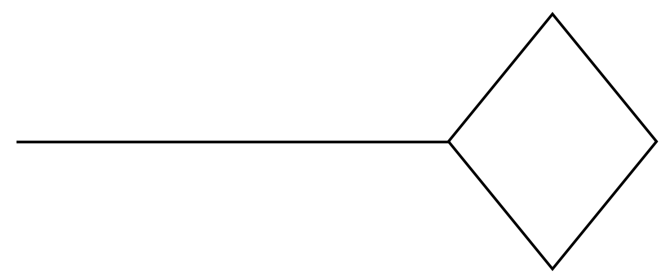

multiplicity
1..*
*
Person
Company
employee
employer
association

Aggregation

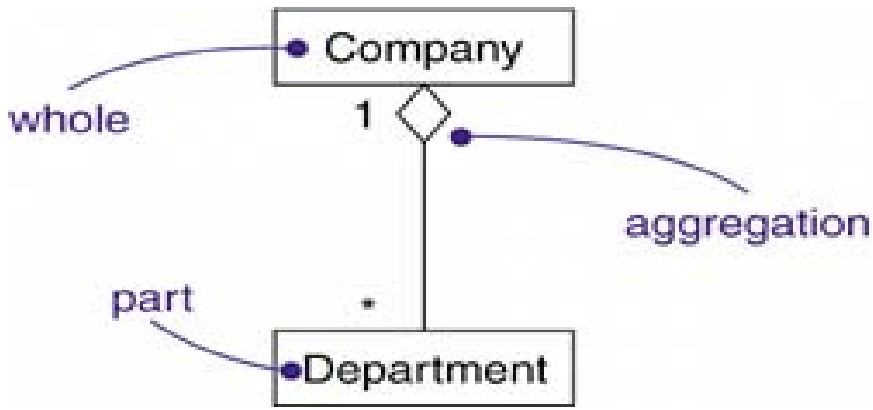
- **Definition:** It models a "whole/part" relationship, in which one class represents a larger thing (the "whole"), which consists of smaller things (the "parts").

- represents a "has-a" relationship, meaning that an object of the whole has objects of the part.

- Aggregation is really just a special kind of association and is specified by adorning a plain association with an unfilled diamond at the whole end,
- Notation:



Example

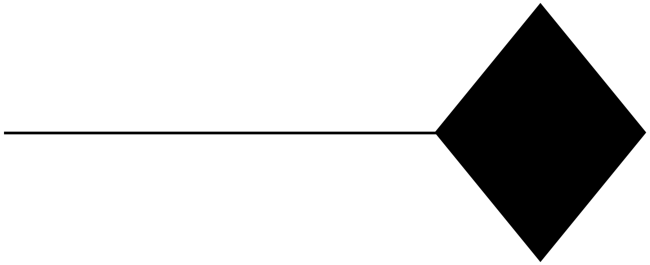


Composition

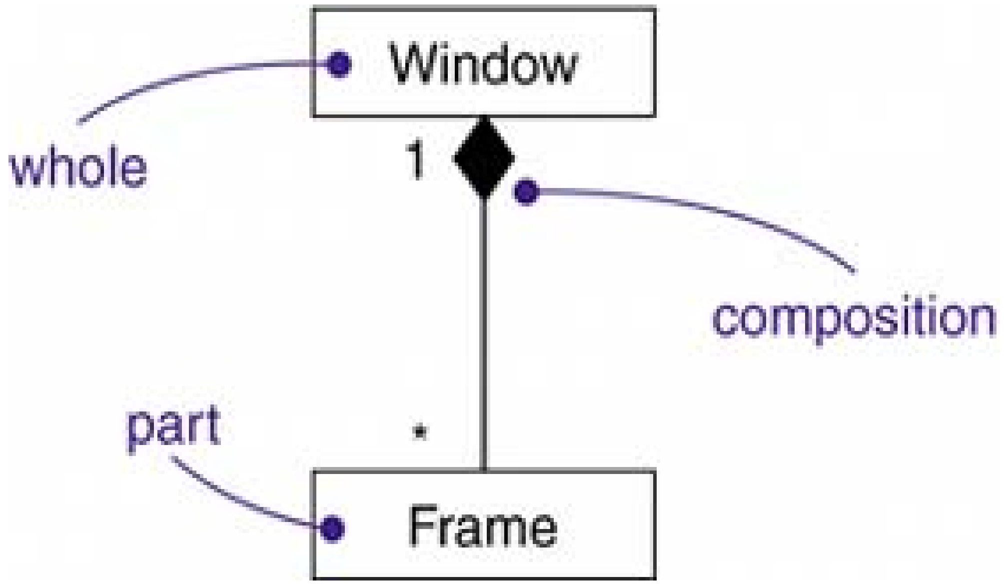
- **Definition** : Composition is a form of aggregation, with strong ownership and coincident lifetime as part of the whole.

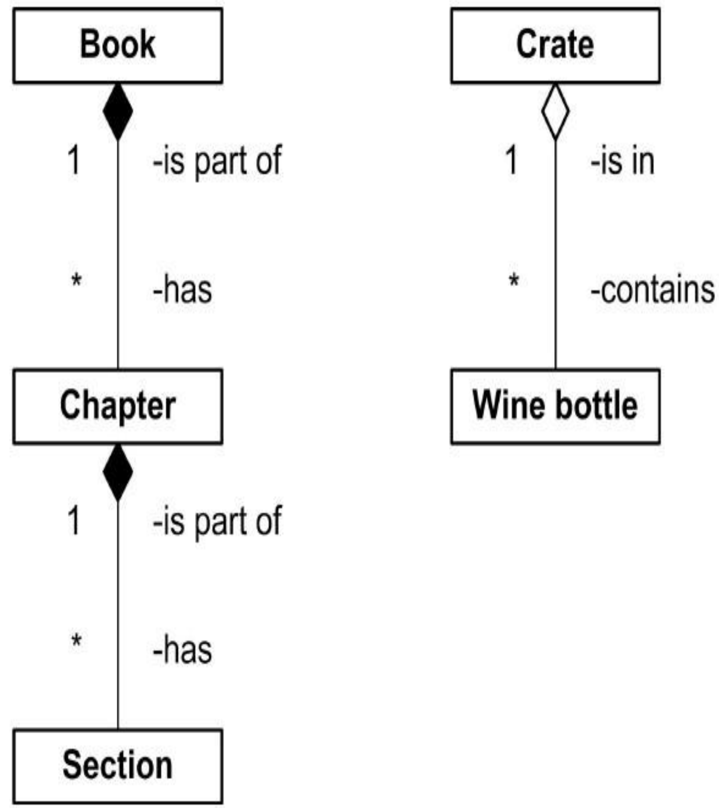
  Existence of component object may depend on the existence of aggregate object.

- Composition is really just a special kind of association and is specified by adorning a plain association with an filled diamond at the whole end,

- Notation: 

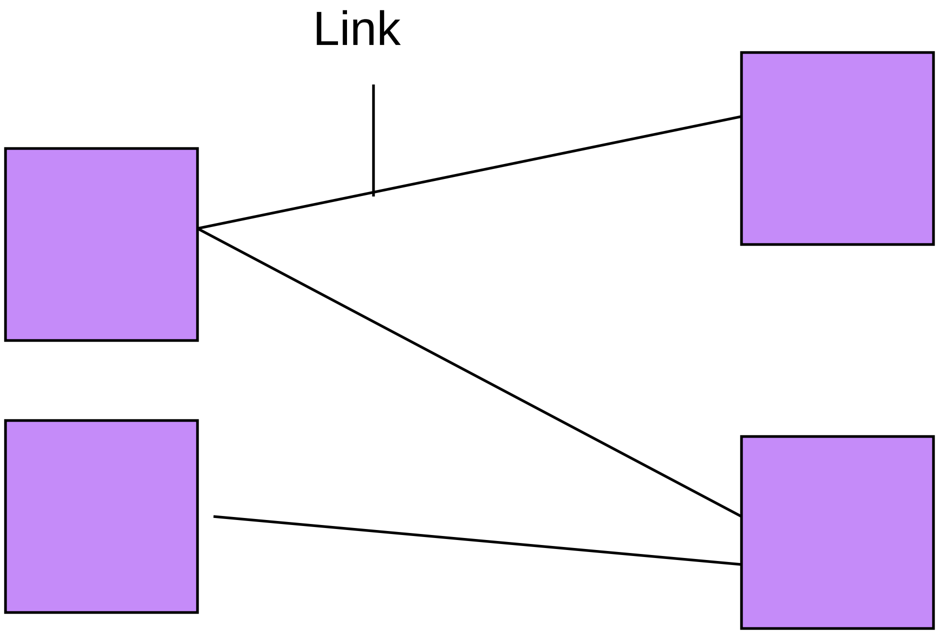
Examples:



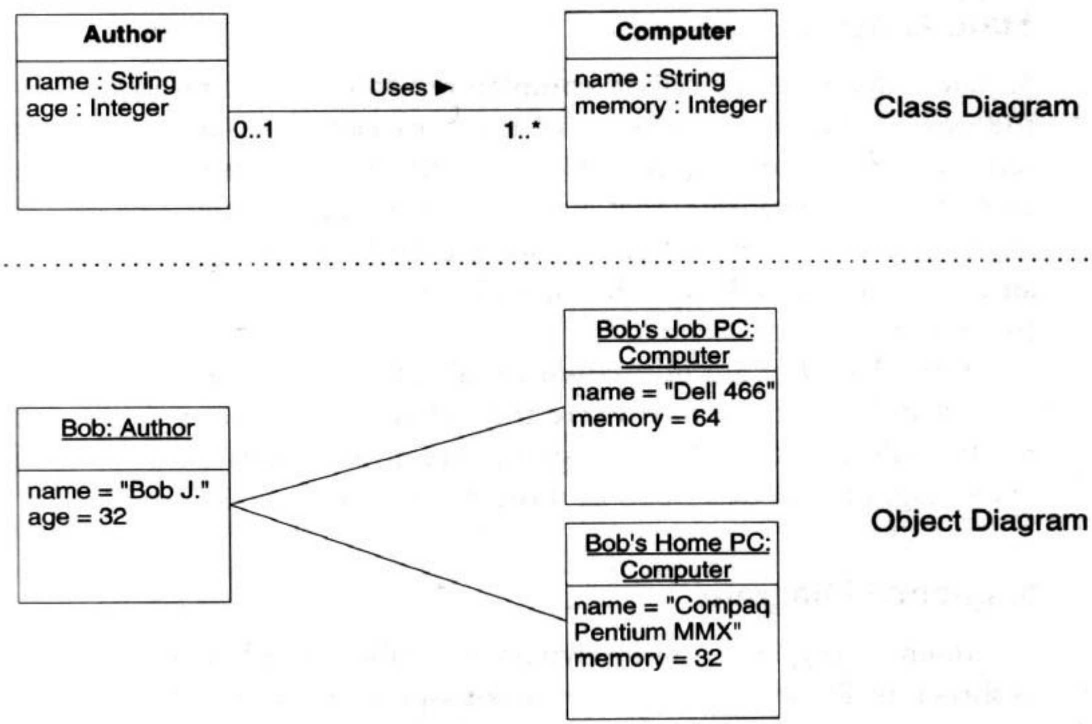

Book
1
-is part of
*
-has
Chapter
1
-is part of
*
-has
Section
Crate
1
-is in
*
-contains
Wine bottle

Link

- Connection among objects.
- Is an instance of association.



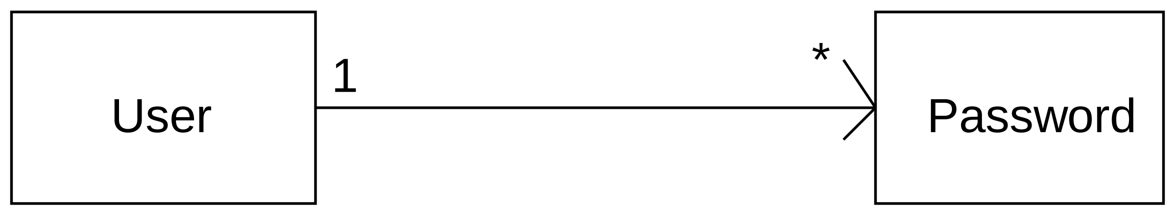
Example


Author
name : String
age : Integer
Uses ▶
0..1
1..*
Computer
name : String
memory : Integer
Class Diagram
Bob: Author
name = "Bob J."
age = 32
Bob's Job PC: Computer
name = "Dell 466"
memory = 64
Bob's Home PC: Computer
name = "Compaq Pentium MMX"
memory = 32
Object Diagram

Navigation

- Unless otherwise specified, navigation across an association is bidirectional.
- However, there are some circumstances in which you'll want to limit navigation to just one direction.
- Notation: →

Example

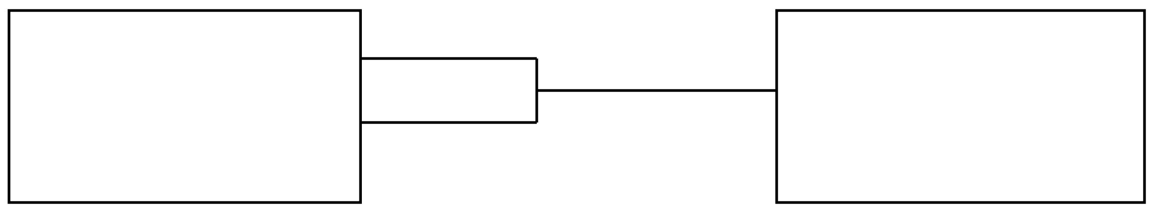


Qualification (Qualified Association)

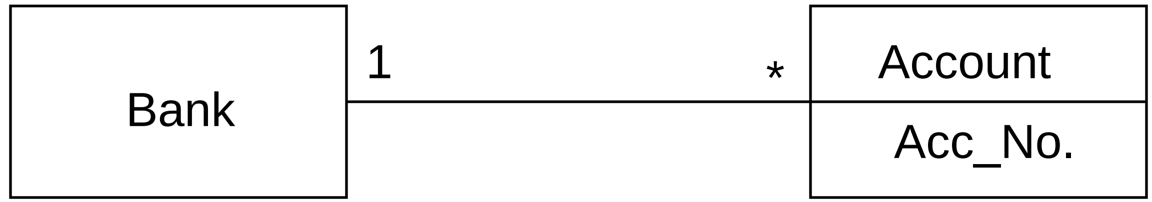
- It relates two object classes and a <u>qualifier.</u>
- Qualifier is a special attribute that reduces effective multiplicity of an association.
- 1..* and *..* may be qualified.

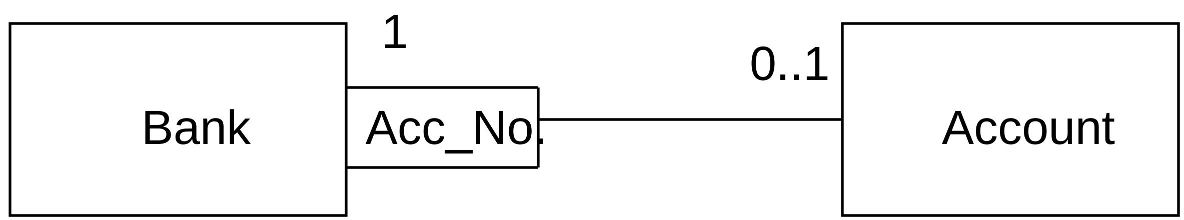
- You render a qualifier as a small rectangle attached to the end of an association, placing the attributes in the rectangle.

Notation and Examples



Source class + Qualifier = target class





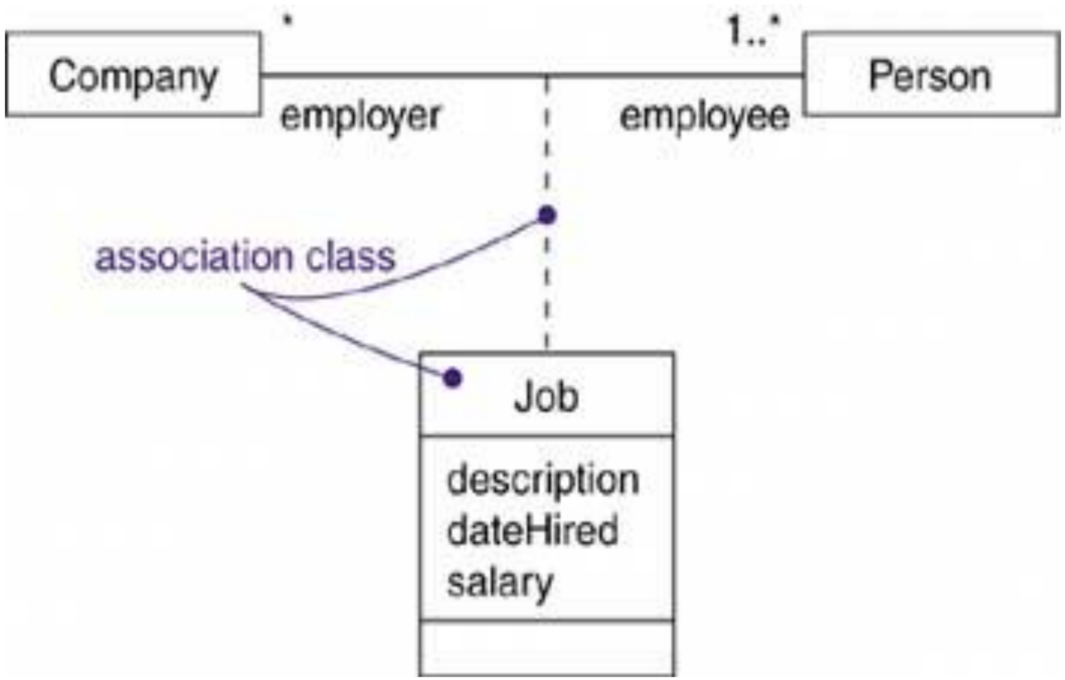
Take another example of company and employee

Association class

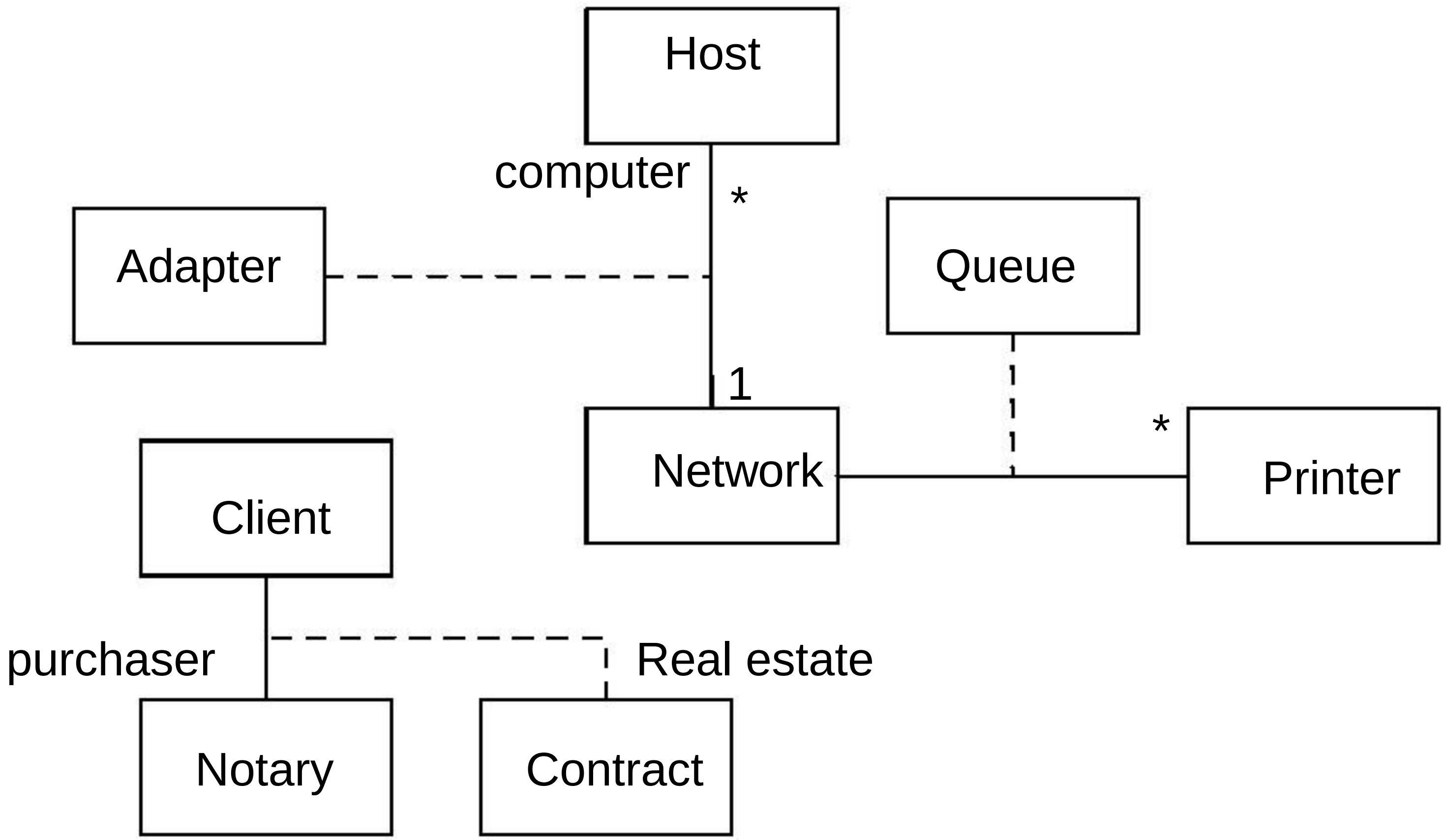
- It is an association that is also an class.
- In an association between two classes, the association itself might have properties.
- For example, in employer/employee relationship between a Company and a Person, there is a Job that represents the properties of that relationship that apply to exactly one pairing of the Person and Company.

- An association class can be seen as an association that also has class properties or as a class that also has association properties.
- You render an association class as a class symbol attached by a dashed line to an association line

Examples


*
1..*
Company
Person
employer
employee
association class
Job
description
dateHired
salary

Examples of Association Class



University Questions

- Explain following with suitable examples:
  - multiplicity, generalization, aggregation, qualified association.
- Write short notes on:
  - Association and Aggregation
  - Multiplicity and Generalization
- Explain relationships of all Object Oriented Models with a diagram. [May 2015]